

Doctoral Research Statement of Intent

Topologically Grounded Diffusion and Causal Spatial Representations for Bidirectional Sign Language AI

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Proposed Primary Supervisor: Dr. Azade Farshad · Recruitment Call: Autumn 2026 ELLIS Doctoral Fellowship

Live 40-DOF Prototype: [WebGL Demo](#) | Full 14-Page Proposal: [Direct PDF](#) | Code: [GitHub Repo](#)

EXECUTIVE RESEARCH VISION: SYSTEMIC OBJECTIVES & ARCHITECTURE

- **Systemic Objective:** An autonomous, real-time (< 100 ms latency) bidirectional assistive communication digital twin enabling spontaneous cross-modal dialogue between Deaf native signers and spoken-language professionals in clinical and civic domains.
- **Tripartite Foundation Architecture:** (1) **Articulatory Perception Channel:** Ingests multi-channel signing video at web-scale (YouTube/broadcast archives) to discretize continuous video into structured linguistic tokens. (2) **Multimodal Foundation Reasoning Core:** Adapts an open-weight frontier model (**Llama 3.1 8B**) via vocabulary expansion ($\mathcal{V}_{\text{extended}} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{sign}}$) on EuroHPC LUMI to internalize 3D spatial syntax and persistent conversation memory (M_t). (3) **Embodied Biomechanical Actuator:** Deterministically compiles predicted tokens into continuous 60 Hz 3D avatar motion on WebGL under anatomical joint limits and strain optimization.
- **Execution & Governance:** 4-year cumulative PhD (ECCV $\rightarrow$ EMNLP $\rightarrow$ ICLR) on CSC's **EuroHPC LUMI supercomputer**, co-designed with native Deaf signers at the Finnish Association of the Deaf (*Kuurojen Liitto ry*).

1 Systemic Objective, Problem Formulation & Societal Impetus

Over **70 million Deaf individuals** worldwide communicate primarily through natural sign languages [4, 13]. In critical civic, educational, and healthcare environments, spontaneous interaction between Deaf signers and hearing professionals is severely obstructed by the structural scarcity of on-demand qualified human interpreters. Under the World Health Organization (WHO) definition of health as a state of comprehensive physical, mental, and social well-being, this communicative barrier represents a profound determinant of social exclusion and medical vulnerability. In Finland and the European Union, equal access is legally guaranteed under Finland's Sign Language Act (*Viittomakielilaki 359/2015*) and the European Accessibility Act (Directive 2019/882).

Computational Bottlenecks in Contemporary Sign Language AI: Unlike spoken natural languages that benefit from trillions of digitized text tokens, **sign languages possess no native written orthography**—they exist as multi-channel, asynchronous 4D spatio-temporal broadcasts across physical space [4]: coordinating 40-DOF dual-hand dexterity, facial Action Units (eyebrow raises AU1/AU2/AU4), and virtual 3D reference loci in the air (*R*-loci) [10]. Prior machine learning paradigms suffer from two fundamental failure modes:

- **Generative Synthesis Artifacts:** Unconstrained neural generative models lack anatomical bone constraints, inducing non-physiological joint dislocations and phalangeal stretching that smear delicate phonological minimal pairs (e.g., confusing MOTHER at the chin with FATHER at the forehead).
- **Perceptual Spatial State Collapse:** Streaming translation models operating on truncated temporal sliding windows purge historical spatial loci, causing models to lose track of discourse referents and corrupting directional agreement verbs like INVITE or VISIT.

This doctoral research overcomes both bottlenecks by coupling deterministic biomechanical kinematics with an adapted multimodal foundation model.

2 Proposed Technical Methodology

2.1 1. Forward Pass: Procedural Scaffolding with Continuous Residual Diffusion

Rather than forcing a neural denoiser to hallucinate high-dimensional 3D kinematics from scratch under global data scarcity (~ 80 hours), we decouple symbolic syntax from physical naturalization across three stages:

1. **Symbolic Syntax & Spatial Allocation:** Spoken text is parsed into sign Time-Topic-Comment (TTC) order. Semantic entities are assigned 3D coordinate anchors (Locus_1 at the sternum, $\text{Locus}_{3a}/\text{Locus}_{3b}$ laterally).
2. **Deterministic Procedural Scaffolding:** Formulated with academic mentorship from Dr. Marco Raoul Marini (Sapienza University of Rome) and empirical lexical corpora (ASL-LEX 2.0 [2, 12], 3D-LEX v1.0 [11]), signs are assembled from atomic sub-lexical primitives. A deterministic 52-joint SMPL-X scaffold $\hat{\mathbf{P}} \in \mathbb{R}^{T \times 52 \times 3}$ is generated via 3-phalanx digit kinematic chains and analytical 2-bone arm Inverse Kinematics (IK) with circular swivel-plane projection and a singularity escape regularizer ($\|\hat{v}_{\text{swivel}} \times \vec{u}\| + \epsilon$).
3. **Continuous Residual Diffusion:** The scaffold provides a continuous geometric prior ($\mathbf{X}_{\text{final}} = \hat{\mathbf{P}} + \Delta\mathbf{X}$). A Temporal Diffusion Transformer (DiT) denoises residual micro-displacements $\Delta\mathbf{X}$, governed by our geometry-preserving objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diffusion}} + \lambda_{\text{geom}} (\mathcal{L}_{\text{landmark}} + \mathcal{L}_{\text{manifold}}) \quad (1)$$

where $\mathcal{L}_{\text{landmark}}$ enforces localized spatial tolerance bubbles ($r_{\text{tol}} \leq 5$ mm) to protect minimal pairs; and $\mathcal{L}_{\text{manifold}}$ enforces bone-length graph invariance and anatomical limits (extending SCOPE [6]), suspended during contact ($\gamma_{\text{contact}}(t)$) for crisp emphatic taps.

2.2 2. Backward Pass: Dual-Rate Perception and Causal Spatial Memory (M_t)

1. **Dual-Rate Stream & Active Visual Saccades:** We decouple continuous processing into a 60 Hz kinematic stream (15-DOF finger angles) and a 1 Hz vision encoder extracting scene context ($\mathbf{z}_{\text{scene}}$) from monocular RGB video (with active depth / LiDAR suggested as an edge-client extension). When decoding entropy spikes, an active visual saccade fetches a localized crop ($\mathbf{v}_{\text{patch}}$), resolving classifier ambiguity (e.g., CL:C) with $> 95\%$ compute savings.
2. **Causal Spatial Memory Bank (M_t):** An external stateful register $M_t = \{\langle \mathbf{x}_k, \mathbf{e}_k, \alpha_k, \tau_k \rangle\}_{k=1}^K$ tracks active 3D referents in Euclidean space. Bound via convergent gaze and manual pointing vectors, active loci decay exponentially. When directional agreement verbs occur, spatial θ -role attention over M_t resolves Source and Goal loci for path verbs, while a normalized aiming vector $\vec{v}_{\text{aim}}$ resolves non-displacement orientation verbs (LOOK-AT, WARN). A verb-class mapping $\mathcal{M}(v)$ [9] maps them to syntactic arguments, resolving regular, backward (INVITE), and orientation agreement immune to temporal window truncation.

2.3 3. The BiSign-LLM: Unified Multimodal Foundation Modeling

Drawing on embodied AI foundations (e.g., Google RT-2 [1], PaLM-E, SpatialVLM), bidirectional coupling is unified via the **BiSign-LLM**. Crucially, rather than training an 8B Transformer from scratch without world knowledge, we adapt an open-weight frontier foundation model (Llama 3.1 8B / Mistral NeMo 12B) via **Vocabulary Expansion** ($\mathcal{V}_{\text{extended}} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{sign}}$). The model undergoes continual self-supervised pre-training on EuroHPC LUMI over millions of sign token streams harvested at web-scale from in-the-wild videos (YouTube, Instagram) by Module 1 (Perceptual Tokenizer). In the forward direction, the LLM emits 1–2 Hz macro-tokens compiled by Module 2 (our 40-DOF dual-hand actuator) into continuous 60 Hz 3D animation; in the backward direction, perceptual tokens from M_t decode into fluent text without context collapse.

3 Methodological Synergy with Dr. Azade Farshad's Group & ELLIS Finland

While Dr. Farshad's group has established international leadership in medical image analysis and surgical data science, my research adapts this geometric and generative toolkit to an urgent, complementary domain: **continuous sign language AI as an assistive communication digital twin**. This initiative directly aligns with ELLIS Institute Finland's core strategic priority, **AI for Health & Society**, addressing communication equity as a vital determinant of mental health and social inclusion.

Rather than viewing sign language through a generic NLP lens, this proposal builds directly upon three foundational pillars of Dr. Farshad's research:

1. **Inference-Time Scene-Graph & Spatial Guidance (Extending SceneGenie [5]):** In *SceneGenie*, Dr. Farshad introduced inference-time sampling guidance using structured scene graphs and augmented bounding gradients ($\nabla \mathcal{L}_{\text{aug,boxg}}$). In sign language, discourse syntax operates as an evolving 3D scene graph $\mathcal{G} = (\mathcal{V}_{\text{loci}}, \mathcal{E}_{\text{predicates}})$. Adapting her training-free spatial guidance enables our residual diffusion denoiser to steer hand effectors toward 3D anatomical loci without spatial drift, while grounding physical environment objects in deictic pointing.
2. **Topological Continuity Preservation (Extending SCOPE [6]):** In medical image analysis, Dr. Farshad demonstrated that biological structures require explicit graph continuity regularizers (c1Dice, Betti numbers β_0, β_1) to prevent severed anatomical components. Articulated human hands exhibit identical topological vulnerabilities during neural diffusion (severed phalanges, rubber digits). Adapting *SCOPE*'s centerline continuity principles guarantees skeletal bone-length invariance ($\mathcal{L}_{\text{graph}}$) and physiological strain compliance (P_{strain}).
3. **Hierarchy-Aware Multimodal Generation (Extending HieraSurg [7]):** In clinical video generation, Dr. Farshad introduced a two-stage hierarchical diffusion framework where high-level procedural states (phases/triplets) synthesize coarse structural maps that condition fine-grained continuous video synthesis. We translate this hierarchical paradigm directly to our Dual-Rate Streaming Architecture: discrete phonological macro-tokens S at 1–2 Hz establish coarse kinematic scaffolds, which condition 60 Hz continuous residual motion diffusion.

4 Phased Research Plan: Core Scope vs. Collaborative Horizons

To maintain a rigorous, realistic doctoral scope, the 4-year research plan is organized around a **3-Paper Cumulative Doctoral Strategy**, distinguishing core deliverables from collaborative stretch horizons:

Phase / Timeline	Core Focus Area	Key Research Activities & Target Venues
Years 1–2(Parallel Tracks A & B)	Kinematic Actuator & Web Tokenizer	- Ethical approval from Aalto Ethics Committee; convene Deaf Advisory Panel with <i>Kuurojen Liitto ry</i>. - <i>Track A (Actuator)</i>: 40-DOF dual-hand procedural kinematics (30 joints) with Battison symmetry/dominance; Continuous Residual Diffusion with localized tolerance ($r_{tol} \leq 5\text{ mm}$). - <i>Track B (Tokenizer)</i>: Web-scale video harvesting (YouTube/IG), HaMeR hand tracking, and $\mathcal{O}(1)$ Causal Spatial Memory (M_t) with deictic parity inversion ($\mathbf{P}^{deictic}$). - Paper 1 Deliverable (Month 17, March 2028): <i>Anatomically Constrained Residual Diffusion for Articulated Sign Kinematics</i>. Target: ECCV 2028.
Year 3(Core Foundation)	BiSign-LLM Foundation Adaptation	- Adapt open-weight foundation model (Llama 3.1 8B / Mistral NeMo) via vocabulary expansion $\mathcal{V}_{extended}$. - Continually pre-train on EuroHPC LUMI (AMD MI250X nodes) across web-harvested sign token streams; execute bidirectional instruction tuning (Sign $\leftrightarrow$ Text). - Paper 2 Deliverable (Month 33, June 2029): <i>BiSign-LLM: Foundation Model Adaptation for Continuous Sign AI</i>. Target: EMNLP 2029.
Year 4(Collaborative Horizons)	Unified Foundation Model & Open-Domain Chat	- Couple BiSign-LLM with residual diffusion for bidirectional interaction across ASL and Finnish Sign Language (SVK). - Benchmark on an Open-Domain Everyday Conversational Study with certified Deaf/SVK consultants from <i>Kuurojen Liitto ry</i>. - Paper 3 Deliverable (Month 36, September 2029): <i>Multimodal Spatial Foundation Modeling for Open-Domain Assistive Communication</i>. Target: ICLR 2030. - Statutory Pre-Examination (<i>Esitarkastus</i>, M42–M47) and Public Defense (<i>Väitöstilaisuus</i>, M48) at Aalto University.

Table 1: Phased Doctoral Work Plan organized into a 3-Paper Cumulative Dissertation Structure.

4.1 Computational Feasibility & Ethical Governance

EuroHPC LUMI Supercomputer: Large-scale pretraining on Motion-X++ (19.5M frames) [16] will utilize CSC’s LUMI supercomputer in Kajaani, Finland. Training will run on LUMI-G AMD Instinct MI250X nodes via native PyTorch AMD ROCm / HIP and high-speed parallel Lustre scratch storage.

Participatory Co-Design: In adherence to “Nothing About Us Without Us,” an ongoing **Deaf Advisory Panel** will be formed with the Finnish Association of the Deaf (*Kuurojen Liitto ry*) from Month 1, providing qualitative guidance on avatar facial blendshapes and linguistic naturalism.

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